

Chapter 4, Problem 2RQ

Problem

For superposition, it is not required that only one independent source be considered at a time; any number of independent sources may be considered simultaneously.

- (a) True
- (b) False

Step-by-step solution

Step 1 of 2

The principle of superposition helps us to analyze a linear circuit with more than one independent source by calculating the contribution of each independent source separately.

Step 2 of 2

However, to apply superposition principle, consider one independent source at a time while all other independent sources are turned off. This implies that replace every voltage source by 0 V (or a short circuit), and every current source by a 0 A (or open circuit).

Therefore, the statement is true.

Therefore, the correct option is ☒ (a) .